
Stabilized Lottery Tickets for Robust Deep Hedging under Transaction Costs

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Abstract

We study sparse deep hedging under transaction costs and stochastic volatility. Using magnitude pruning and stabilized lottery tickets via late rewinding (epoch 30), we evaluate both clean hedging performance and robustness under adversarial feature perturbations (FGSM/PGD) and distribution shifts (regime changes). We further build robust sparse hedging policies using an efficient FGSM→PGD adversarial training procedure, and provide diagnostics to interpret which features and network components drive robustness.

1 Introduction

Deep hedging learns dynamic trading strategies to hedge derivative payoffs under market frictions [Buehler et al., 2019]. In realistic settings with transaction costs and stochastic volatility, optimal hedging policies are non-linear and path-dependent, and analytical solutions are generally unavailable or impractical. Neural networks provide a flexible class of policies: trained on simulated trajectories, they map available market information at each hedging date to a position in the underlying, and are optimized end-to-end for a risk objective defined on the terminal hedging P&L.

This project investigates whether deep hedging can be made more efficient and reliable by leveraging *sparsity* and *robustness*. First, dense networks may be costly to train/deploy; the *Lottery Ticket Hypothesis (LTH)* [Frankle and Carbin, 2019] suggests that a sparse subnetwork (a “ticket”) can match the dense model’s performance when trained appropriately. Second, hedging strategies are exposed to stress: small input misspecifications, noisy signals, or distribution shifts (regime changes) may induce large losses. We therefore evaluate both standard performance and robustness, and study robust training procedures for sparse hedgers, including adversarial stress tests and adversarial training [Goodfellow et al., 2015, Madry et al., 2018, Ilyas et al., 2019].

Main findings. Our main findings are that:

- (a) stable lottery tickets for deep hedging require *late rewinding* (we use epoch 30),

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- (b) sparse tickets can match the dense baseline at moderate sparsity, but are vulnerable to adversarial stress unless trained robustly, and
- (c) an efficient FGSM→PGD adversarial training procedure yields robust sparse hedging policies with competitive performance.

Contributions.

- (1) A reproducible deep hedging pipeline under transaction costs with strong baselines and sanity checks.
- (2) An LTH study for hedging, including controlled comparisons separating the role of the pruning mask from the role of the initialization/rewinding.
- (3) A robust-training recipe (FGSM→PGD) and an empirical robustness evaluation under adversarial attacks and regime shifts, complemented with interpretability analyses.

2 Problem Setup: Deep Hedging under Transaction Costs

2.1 Market model, payoff, and simulation scheme

We consider a discrete-time hedging problem over dates $t_0 = 0 < t_1 < \dots < t_T = T$. At each date, the hedger chooses a position δ_t in the underlying, and the resulting strategy is evaluated through its terminal P&L, including transaction costs and the option payoff. The underlying price $(S_t)_{t \in [0, T]}$ follows a stochastic-volatility model of Heston type under the risk-neutral measure [Heston, 1993],

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^S, \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (dW_t^S, dW_t^v) = \rho dt, \quad (2)$$

where r is the risk-free rate and v_t is the instantaneous variance. We focus on a short European call option with strike K and payoff

$$Z = (S_T - K)^+. \quad (3)$$

Simulation scheme (Inverse Gaussian). To simulate the Heston variance process robustly, we rely on an Inverse Gaussian (IG) discretization scheme for the square-root (CIR-type) variance dynamics, following Abi Jaber [2024]. This choice avoids negative variance artifacts that can arise with naive Euler-type discretizations and provides stable simulations at the time-step size used in our hedging grid.

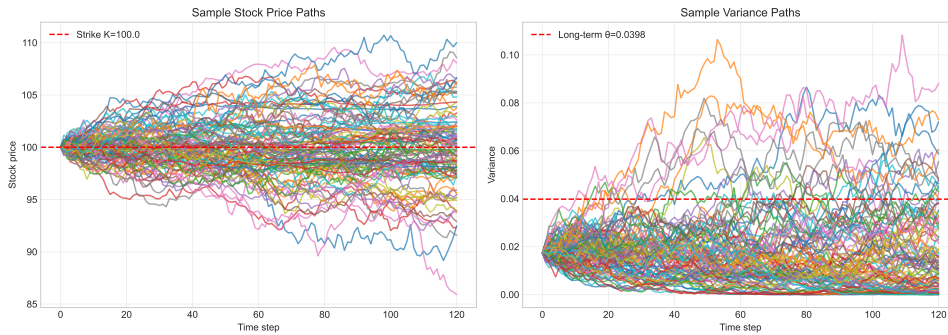


Figure 1: Sample simulated Heston trajectories for the underlying and variance processes.

Data generation rationale. We rely on simulated data following standard practice in the deep hedging literature [Buehler et al., 2019]. Real options market data (e.g., from CBOE or Bloomberg) requires costly subscriptions and presents additional challenges including asynchronous quotes, illiquidity, and unobserved true volatility. Simulation provides controlled experiments where ground truth is known, enabling rigorous evaluation of hedging performance. This approach is adopted universally in foundational deep hedging work and subsequent extensions.

2.2 Dynamic hedging strategy

A trading strategy is a sequence $(\delta_t)_{t=0}^{T-1}$, where δ_t denotes the number of shares of the underlying held over $(t, t_{t+1}]$. The strategy is implemented dynamically: at each time step, the hedger observes the market state and updates the position from δ_{t-1} to δ_t .

2.3 P&L with proportional transaction costs

Let $\Delta S_t = S_t - S_{t-1}$. In our implementation, the terminal hedging profit and loss (P&L) for the short call is

$$\text{PnL} = \sum_{t=1}^T \delta_{t-1} \Delta S_t - Z - \text{TC}, \quad (4)$$

where TC denotes proportional transaction costs incurred when changing the position:

$$\text{TC} = c_{\text{prop}} \left(|\delta_0| S_0 + \sum_{t=1}^{T-1} |\delta_t - \delta_{t-1}| S_t + |\delta_{T-1}| S_T \right), \quad (5)$$

2.4 Risk objective (OCE/CVaR) and learned premium

Rather than optimizing only average performance, we emphasize tail risk. For a confidence level $1 - \alpha$, CVaR measures the *average loss in the worst α fraction of outcomes* [Rockafellar and Uryasev, 2000]; here it corresponds to the average shortfall in the worst market scenarios. We optimize a CVaR-type objective through an OCE formulation with an auxiliary scalar $y \in \mathbb{R}$ learned jointly with the policy. Importantly, y is not injected into the P&L in (4); it only appears in the OCE objective. For a tail level $\alpha \in (0, 1)$, the loss takes the form

$$\mathcal{L}(\delta, y) = y + \frac{1}{\alpha} \mathbb{E}[\max(-\text{PnL} - y, 0)], \quad (6)$$

which corresponds to optimizing a CVaR-type risk measure of the hedging shortfall [Rockafellar and Uryasev, 2000].

2.5 Inputs to the hedging policy

The hedging policy is parameterized by a neural network that outputs δ_t at each time step. At time t , the network receives (i) a vector of exogenous market features (e.g., log-moneyness, returns, volatility proxies, and time-to-maturity), and (ii) recurrent state variables summarizing the previous hedge and accumulated trading activity (e.g., δ_{t-1} , cumulative P&L, and cumulative trading volume). This explicit recurrence enforces a consistent trading dynamics and allows transaction costs to meaningfully shape the learned policy [Buehler et al., 2019].

3 Methodology and Experimental Design

This section describes the end-to-end pipeline used to (i) train a deep hedging policy under transaction costs, (ii) extract sparse subnetworks via pruning and late rewinding, and (iii) evaluate and improve robustness under adversarial stress tests and distribution shifts.

3.1 End-to-end deep hedging pipeline

Our workflow is organized as a sequence of differentiable blocks: we simulate market paths, compute a feature representation, apply a neural policy that outputs trading positions, execute self-financing trading with transaction costs, and finally optimize a tail-risk objective on the resulting terminal P&L, as in Buehler et al. [2019].

3.2 Sparse subnetworks via pruning and stabilized lottery tickets

We study whether sparsity can be exploited in deep hedging through the Lottery Ticket Hypothesis (LTH) [Frankle and Carbin, 2019]: starting from a trained dense network, we prune weights to obtain a binary mask and retrain the remaining subnetwork. We use unstructured magnitude pruning (global across prunable layers) and keep the induced sparse structure fixed during retraining.

Late rewinding (stabilized tickets). In our setting, rewinding the surviving weights back to their random initialization proved unstable. We therefore adopt *late rewinding*: after pruning, surviving weights are reset to their values at an intermediate checkpoint (we use epoch 30 throughout), following the stabilization protocol of Frankle et al. [2020]. This defines the stabilized-ticket procedure used in the rest of the project.

Controlled comparisons (mask vs. initialization). To disentangle the role of the pruning mask from the role of the weights used for retraining, we include two key controls:

1. **Random mask + random initialization:** a sparsity-matched model without inherited structure.
2. **Fixed mask + random initialization:** reuse the mask extracted from the dense baseline, but reinitialize the remaining weights, to test whether performance comes from the mask structure alone or from the rewound weights.

3.3 Adversarial stress tests on the feature space

Robustness matters in hedging because small misspecifications in inputs (volatility proxy errors, microstructure noise, stale signals, etc.) can produce large changes in the trading rule and therefore in tail losses. We evaluate robustness through *adversarial stress tests* [Goodfellow et al., 2015, Madry et al., 2018, Ilyas et al., 2019], which approximate a local worst-case perturbation of the model input.

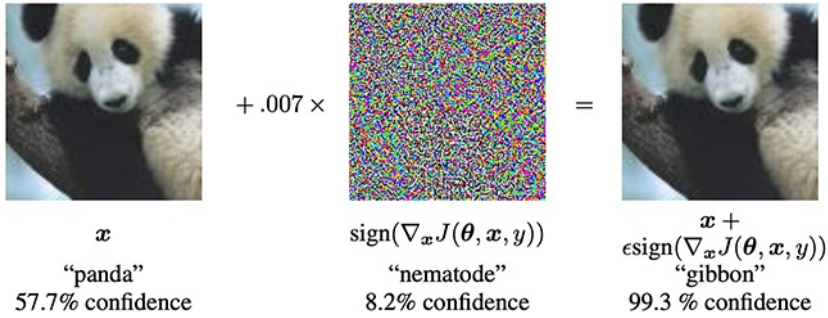


Figure 2: Adversarial perturbations: an imperceptible input change can induce a large model-output change. We use this concept to stress-test hedging policies by perturbing their input features.

Threat model in our setting. Unlike image classification, our inputs are structured financial features. We keep the simulated market trajectory (and therefore the payoff) fixed, and perturb only the feature tensor provided to the policy network. This isolates sensitivity to *signal* misspecification while keeping a well-defined underlying path.

Concretely, we perturb only the five exogenous market features (log-moneyness, returns, volatility, variance change, time-to-maturity). The three recurrent state variables (δ_{prev} , cumulative P&L, trading volume) are computed internally by the model during the forward pass and are therefore not subject to adversarial perturbation.

FGSM and PGD. We consider additive perturbations Δ constrained in an uncertainty set $\mathcal{U}$ (typically an ℓ_∞ -ball): $x^{\text{adv}} = x + \Delta$ with $\|\Delta\|_\infty \leq \varepsilon$. For *evaluation*, we report the worst-case degradation induced by solving (approximately)

$$\max_{\Delta \in \mathcal{U}} \mathcal{L}(\pi_\theta(x + \Delta), y). \quad (7)$$

We use two standard white-box attacks: *FGSM* performs a single step in the direction of the sign gradient of the loss with respect to the inputs [Goodfellow et al., 2015], while *PGD* iterates FGSM steps with projection back onto $\mathcal{U}$ [Madry et al., 2018]. PGD is stronger but more expensive, making it a natural tool both for evaluation and for robust training.

3.4 Adversarial training: FGSM $\rightarrow$ PGD curriculum

Adversarial training improves robustness by optimizing the policy under adversarially perturbed inputs [Madry et al., 2018]. Formally, it corresponds to the min-max objective

$$\min_{\theta} \mathbb{E} \left[\max_{\Delta \in \mathcal{U}} \mathcal{L}(\pi_{\theta}(x + \Delta), y) \right]. \quad (8)$$

In practice, multi-step PGD inside the inner maximization is costly. We therefore adopt a two-phase procedure: a fast FGSM phase to quickly reach a meaningful region of parameter space, followed by a stronger PGD phase to obtain genuine robustness. This FGSM $\rightarrow$ PGD curriculum is consistent with the view that pruning and adversarial training are complementary [Li et al., 2020b], and is applied consistently across dense and sparse models. We also refer to the stabilized ticket perspective for adversarial training [Li et al., 2020a].

3.5 Distribution shifts via regime changes

Finally, we evaluate out-of-distribution robustness by changing the data-generating process at test time. Concretely, we alter the stochastic-volatility regime (Heston parameters) to emulate shifts in volatility level, mean reversion, or vol-of-vol. This complements adversarial stress tests: adversarial attacks probe local worst-case sensitivity, while regime shifts probe global misspecification.

Table 1: Key experimental settings. Full configuration in Appendix B.

Component	Setting
Market model	Heston stochastic volatility
Option	European call, $S_0 = K = 100$, $T = 1/12$
Transaction costs	10 bps proportional
Network	MLP [256, 256, 128], ReLU
Objective	CVaR _{5%} (OCE formulation)
Sparsity levels	{50%, 80%, 90%, 95%}
Late rewinding	Epoch 30

4 Experiments and Results

4.1 Evaluation metrics

We evaluate hedging strategies using summary statistics of the terminal hedging P&L distribution. **Mean P&L** reflects the average hedging cost (here negative values correspond to a net loss for a short-call hedger after costs), while **Std P&L** measures dispersion and therefore risk. Since hedging is primarily concerned with adverse scenarios, we report **CVaR _{α}** (Conditional Value-at-Risk), defined as the average P&L over the worst α fraction of outcomes: more negative CVaR indicates larger tail losses. Finally, we report a **Sharpe** ratio (mean divided by standard deviation) as a normalized risk-adjusted measure; in our setting, it should be interpreted cautiously as P&L is not an investable return, but it provides a compact signal-to-noise indicator across methods. The negative values observed throughout our experiments are expected: the P&L measures hedging cost (transaction costs and replication error) without including the option premium that would be received when selling the call.

4.2 Dense baseline and benchmark comparison

Before introducing sparsity or robustness, we first validate the dense deep-hedging baseline trained under the stochastic-volatility Heston dynamics. The model is optimized for a convex tail-risk objective (OCE) using simulated paths with proportional transaction costs.

Convergence and stability. Figure 3 shows smooth convergence of the training and validation losses, from about 30 to 3 within twenty epochs, with minimal gap thereafter. The mean P&L stabilizes near -1.73 and the CVaR_{5%} around -3.0 , confirming a stable optimum without overfitting.

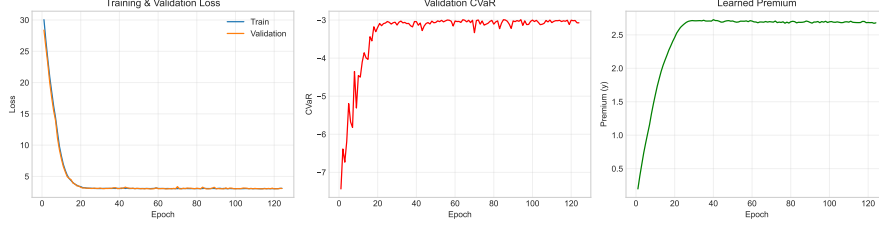


Figure 3: Training and validation loss (OCE) along epochs for the dense baseline.

Performance against benchmarks. The learned policy is compared to several classical hedging rules: delta hedging with constant or instantaneous volatility, static delta positions, and no-hedging. Results in Table 2 and Figure 4 show that deep hedging consistently improves tail-risk metrics.

Method	Mean P&L	Std P&L	CVaR _{5%}	Sharpe
Deep Hedging	-1.73	0.71	-3.00	-2.42
Delta (const. vol)	-1.93	0.66	-3.72	-2.90
Delta (inst. vol)	-1.95	0.56	-3.26	-3.47
No Hedging	-1.58	2.48	-9.23	-0.64
Static $\Delta=0.5$	-1.59	1.30	-5.30	-1.23
Static $\Delta=1.0$	-1.61	2.24	-8.48	-0.72

Table 2: Baseline comparison under the Heston model with transaction costs.

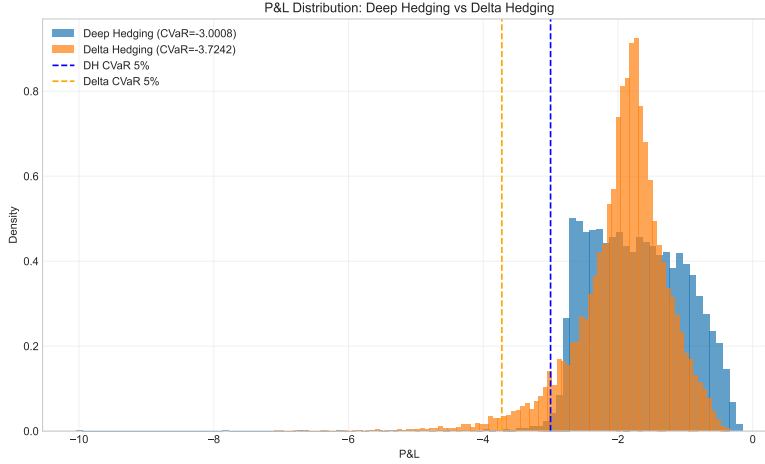


Figure 4: Out-of-sample P&L distribution of the dense deep-hedging policy vs. benchmark strategies.

Interpretation. The deep hedging policy achieves lower tail losses than any benchmark, especially compared to delta hedging calibrated on constant volatility. This validates both the numerical stability of the training pipeline and the relevance of the convex risk objective before introducing sparsity or robustness mechanisms.

4.3 Black-Scholes sanity check (frictionless, constant volatility)

In this part, we validate the correctness of the end-to-end deep hedging implementation in a simplified frictionless setting where classical intuition is clear. We consider a Black-Scholes-like regime obtained as a degenerate Heston configuration: constant variance ($v_t \equiv v_0$) and no transaction costs.

Setup (sanity configuration). We use the same data-generation and training code as in the main pipeline, but set $c_{\text{prop}} = 0$ and freeze the variance dynamics by choosing $\xi \approx 0$.

Baselines (expected reference). In a frictionless constant-volatility regime, delta hedging is a natural benchmark. Since volatility is constant here, `delta_hedging_const_vol` and `delta_hedging_inst_vol` coincide, and provide a reference level of tail risk.

Strategy	Mean P&L	Std P&L	CVaR _{5%}	Sharpe
Delta hedging (const. vol)	-2.2976	0.3607	-3.1309	-6.3692
Delta hedging (inst. vol)	-2.2976	0.3607	-3.1309	-6.3692
No hedging	-2.3516	3.4927	-12.5040	-0.6733
Static $\Delta=0.5$	-2.2781	1.7327	-6.7168	-1.3148
Static $\Delta=1.0$	-2.2045	3.1903	-11.1416	-0.6910

Table 3: Sanity-check baselines in the frictionless constant-volatility regime.

Deep hedging results (test set). We evaluate the trained policy out-of-sample on the test set. The deep hedging policy achieves: $pnl_mean = -2.2962$, $pnl_std = 0.3737$, and $CVaR_{5\%} = -3.1050$. This is very close to the delta hedging benchmark in terms of tail risk and dispersion. We also report tracking-error diagnostics relative to the target replication: $RMSE = 0.0323$, $MAE = 0.0141$, and hedge effectiveness 0.9886.

Metric (test set)	Deep Hedging value
Mean P&L	-2.2962
Std P&L	0.3737
CVaR _{5%}	-3.1050
Sharpe (mean/std)	-6.1450
Tracking RMSE	0.0323
Tracking MAE	0.0141
Hedge effectiveness	0.9886

Table 4: Deep hedging performance on the test set for the Black–Scholes sanity check.

Interpretation (what this sanity check validates). This experiment is a *functional validation*: in a frictionless constant-volatility regime, deep hedging recovers a performance level that matches classical delta hedging in tail-risk terms, while strongly improving upon no-hedging and static deltas. The negative Sharpe values are not an anomaly here: the P&L is defined for a short-call hedger and is negative on average (interpretable as the hedging cost / premium scale), hence mean/std is negative. Overall, the results confirm that the simulator, P&L accounting, risk objective, and optimization pipeline behave as expected in a simplified regime, supporting the credibility of subsequent experiments under stochastic volatility, transaction costs, sparsity, and adversarial stress.

4.4 Sparse subnetworks via lottery tickets

We now investigate whether sparse subnetworks can match the dense baseline’s performance when trained appropriately, following the Lottery Ticket Hypothesis [Frankle and Carbin, 2019].

4.4.1 Pruning protocol and stabilized tickets

We apply global unstructured magnitude pruning to the trained dense network: weights with the smallest absolute values are removed across all layers simultaneously, excluding the output layer to preserve the policy’s expressiveness. Following Frankle et al. [2020], we adopt *late rewinding* rather than resetting to initialization: after pruning, surviving weights are rewound to their values at epoch 30 (denoted θ_{30}), which we found necessary for stable retraining. Rewinding to epoch 0 led to unstable optimization in our setting. The sparse network (approximately 101,000 total parameters when dense) is then retrained with the pruning mask held fixed.

4.4.2 Learning rate sensitivity

The choice of learning rate for retraining pruned networks is critical. Table 5 reports convergence speed and final performance for stabilized tickets at 80% sparsity across different learning rates.

Learning Rate	Epochs to 95%	CVaR _{5%}	Sharpe
10^{-2}	4	-3.04	-2.50
10^{-3}	1	-2.99	-2.52
10^{-4}	4	-2.99	-2.43
10^{-5}	44	-3.03	-2.45

Table 5: Learning rate search for stabilized tickets (80% sparsity). “Epochs to 95%” denotes the number of epochs to reach 95% of baseline validation loss. LR= 10^{-3} offers the best trade-off between convergence speed and final performance.

Lower learning rates yield smoother convergence curves but require more epochs; higher rates converge faster but with increased oscillation. We select LR= 10^{-3} as the best compromise, achieving near-instant convergence (1 epoch to 95% baseline performance) with competitive final metrics. Figure 5 illustrates the convergence dynamics across learning rates. Detailed training curves showing the smoothness-speed trade-off are provided in Appendix A.

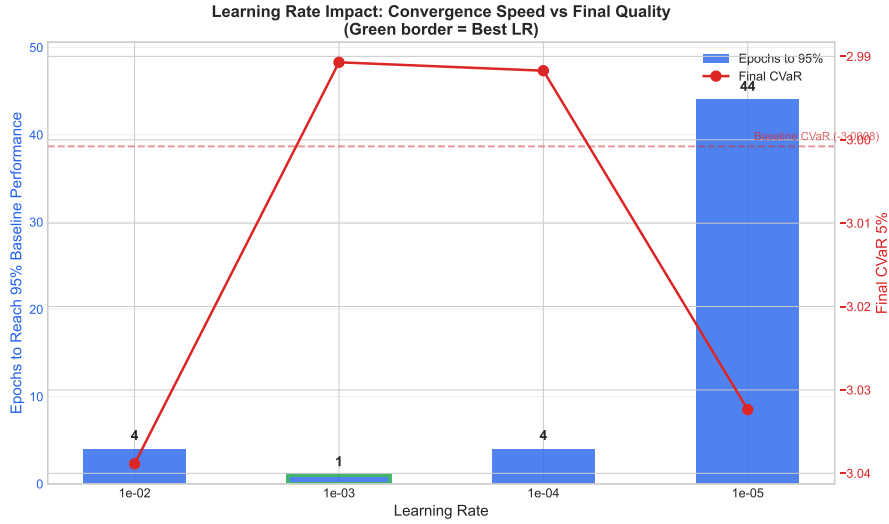


Figure 5: Convergence speed of stabilized tickets (80% sparsity) for different learning rates. LR= 10^{-3} achieves the fastest convergence to baseline performance.

4.4.3 Controlled comparisons: disentangling mask and initialization

A key question is whether performance gains come from the pruning mask structure, the rewind weights, or their combination. We design controlled experiments at 80% sparsity to isolate these factors:

1. **Stabilized ticket:** magnitude-based mask from θ^* + weights rewind to θ_{30} .
2. **Fixed mask + random init:** same mask as (1), but weights reinitialized randomly.
3. **Random mask + random init:** random 80%-sparse mask with random initialization.

Condition	Epochs	CVaR _{5%}	Sharpe	Speedup
Stabilized ticket (mask + θ_{30})	1	-2.99	-2.52	124×
Fixed mask + random init	31	-3.00	-2.43	4×
Random mask + random init	102	-3.08	-2.31	1.2×
Dense baseline	124	-3.00	-2.42	1×

Table 6: Controlled comparisons at 80% sparsity (LR= 10^{-3}). “Epochs” denotes epochs to early stopping or 95% baseline performance. Speedup is relative to dense baseline training time.

The results in Table 6 reveal that: the stabilized ticket converges $124\times$ faster than training the dense network from scratch, while the random mask + random init condition requires nearly as many epochs as the dense baseline (102 vs. 124). This confirms that the random-mask sparse network is *not* a lottery ticket—it behaves like a standard sparse network that must learn from scratch. The fixed mask + random init condition shows intermediate behavior (31 epochs, $4\times$ speedup), indicating that the magnitude-based mask structure provides some benefit, but the *combination* of mask and rewind initialization (θ_{30}) is essential for the dramatic acceleration observed in stabilized tickets.

4.4.4 Performance across sparsity levels

We now evaluate stabilized tickets across sparsity levels from 50% to 95%, using $LR=10^{-3}$ and late rewinding to epoch 30. Table 7 reports test-set performance.

Sparsity	Params	Mean P&L	Std P&L	CVaR _{5%}	Sharpe
0% (dense)	101,122	−1.727	0.713	−3.00	−2.42
50%	~50,500	−1.722	0.718	−3.00	−2.40
80%	~20,200	−1.735	0.701	−3.00	−2.47
90%	~10,100	−1.728	0.703	−2.99	−2.46
95%	~5,050	−1.724	0.716	−3.01	−2.41

Table 7: Stabilized ticket performance across sparsity levels. Even at 95% sparsity (~5,000 remaining parameters), the sparse network matches the dense baseline within 1% on all metrics.

Figure 6 visualizes the trade-off between sparsity and performance. While CVaR and Sharpe ratio remain within 5% of the dense baseline across all sparsity levels, the relationship is *non-monotonic*: the 90% sparse model achieves slightly better CVaR than both the dense baseline and the 95% model. This suggests that moderate sparsity may provide implicit regularization benefits, though the effect is small.

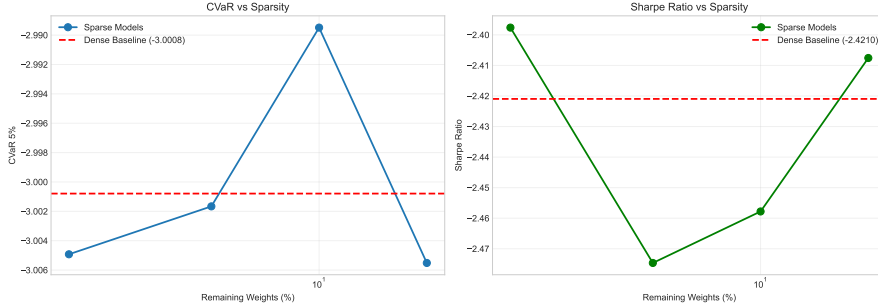


Figure 6: CVaR_{5%} and Sharpe ratio as a function of remaining weights. Stabilized tickets maintain baseline performance up to 95% sparsity.

4.4.5 Summary

Our experiments confirm that the Lottery Ticket Hypothesis extends to deep hedging: sparse sub-networks exist that match dense performance when retrained from appropriate initializations. Three findings are noteworthy: (i) *late rewinding* (epoch 30) is essential for stability; (ii) the combination of magnitude-based mask and rewind weights yields $124\times$ faster convergence than training from scratch, while random masks show no such acceleration; (iii) sparsity up to 95% (retaining only ~5,000 of 101,000 parameters) incurs less than 5% performance degradation. These stabilized tickets provide an efficient foundation for subsequent robustness analysis.

4.5 Adversarial stress tests

We now evaluate the robustness of dense and sparse hedging policies under adversarial perturbations, which approximate worst-case input misspecifications.

4.5.1 Threat model

In hedging applications, input features may be corrupted by estimation errors in volatility proxies, stale price feeds, or microstructure noise. We model such misspecifications as bounded adversarial perturbations applied to the feature tensor x_t provided to the policy network. Crucially, we keep the underlying market trajectory (S_t, v_t) and the option payoff Z fixed; only the features seen by the hedging policy are perturbed. This isolates sensitivity to *signal noise* while maintaining a well-defined hedging problem.

4.5.2 Attack protocols

We consider two standard white-box attacks with perturbations constrained in ℓ_∞ -balls:

FGSM (Fast Gradient Sign Method). A single gradient step in the direction that maximizes the loss [Goodfellow et al., 2015]:

$$x^{\text{adv}} = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L}). \quad (9)$$

We use $\varepsilon_S = 0.02$ (2%) for price-related features and $\varepsilon_v = 0.2$ (20%) for volatility features.

PGD (Projected Gradient Descent). An iterative attack that applies multiple FGSM steps with projection back onto the constraint set [Madry et al., 2018]:

$$x^{(k+1)} = \Pi_{\mathcal{B}_\varepsilon} \left(x^{(k)} + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}) \right), \quad (10)$$

where $\Pi_{\mathcal{B}_\varepsilon}$ projects onto the ε -ball around the original input. We use 10 iterations with $\varepsilon_S = 0.05$, $\varepsilon_v = 0.5$, and step sizes $\alpha_S = 0.01$, $\alpha_v = 0.1$. PGD provides a stronger attack than FGSM at higher computational cost.

4.5.3 Robustness evaluation

Table 8 reports CVaR_{5%} under clean, FGSM, and PGD conditions for the dense baseline and stabilized tickets at various sparsity levels.

Model	Clean CVaR	FGSM CVaR	PGD CVaR	FGSM Gap	PGD Gap
Dense baseline	−3.00	−4.87	−11.81	1.87	8.81
Ticket 50%	−3.00	−5.92	−11.81	2.91	8.80
Ticket 80%	−3.00	−5.73	−11.88	2.73	8.88
Ticket 90%	−2.99	−4.24	−12.96	1.26	9.97
Ticket 95%	−3.01	−5.16	−12.54	2.15	9.54

Table 8: Adversarial robustness evaluation. “Gap” denotes the absolute CVaR degradation from clean to adversarial conditions. PGD attacks induce $\sim 4\times$ larger degradation than FGSM.

Figure 7 visualizes these results. Several observations emerge: (i) all models—dense and sparse—are substantially vulnerable to adversarial perturbations; (ii) PGD attacks are far more effective than FGSM, degrading CVaR by ~ 9 units (from -3 to -12) compared to ~ 2 units for FGSM; (iii) under clean conditions, all tickets match the dense baseline, confirming the results of Section 4.4.

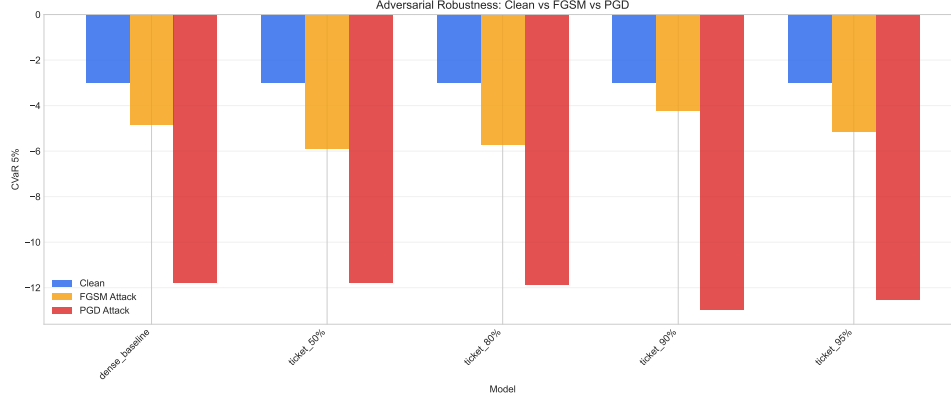


Figure 7: $\text{CVaR}_{5\%}$ under clean (blue), FGSM (orange), and PGD (red) conditions. All models exhibit similar clean performance but substantial degradation under attack, with PGD causing $\sim 4\times$ more damage than FGSM.

4.5.4 Sparsity-robustness relationship

Figure 8 examines whether sparsity amplifies adversarial vulnerability. We plot the PGD CVaR gap (degradation from clean to PGD) as a function of sparsity level.

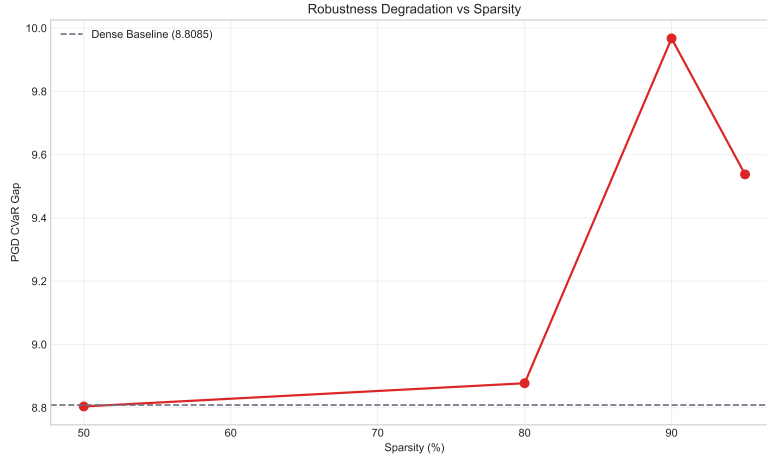


Figure 8: PGD CVaR gap as a function of sparsity. The dashed line indicates the dense baseline gap (8.81). Higher sparsity tends to increase vulnerability, though the relationship is non-monotonic.

The trend suggests that sparsity *does* amplify adversarial vulnerability: the PGD gap increases from 8.80 at 50% sparsity to 9.97 at 90% sparsity (+13% relative increase). However, the relationship is non-monotonic—the 95% ticket (gap 9.54) is slightly more robust than the 90% ticket (gap 9.97). This may reflect the inherent variability in how pruning affects different network structures, or suggest that extreme sparsity forces the network to rely on more stable features. Regardless, all sparse models show degradation at or above the dense baseline level.

4.5.5 Summary

Our adversarial evaluation reveals that both dense and sparse hedging policies are highly vulnerable to input perturbations: PGD attacks degrade CVaR from ~ -3 to ~ -12 , a $4\times$ increase in tail risk. While stabilized tickets achieve comparable clean performance to the dense baseline (Section 4.4), they offer no inherent robustness advantage—and high-sparsity tickets may be slightly more vulnerable. These findings motivate the adversarial training procedure developed in the next section.

4.6 Adversarial training via FGSM→PGD curriculum

Having established that both dense and sparse models are vulnerable to adversarial perturbations, we now develop a training procedure to obtain robust hedging policies. Our approach builds on the boosting ticket framework [Li et al., 2020b], which shows that weak adversarial training can accelerate strong adversarial training.

4.6.1 Motivation and protocol

Adversarial training with PGD [Madry et al., 2018] is effective but computationally expensive: each training step requires multiple forward-backward passes to generate adversarial examples. For deep hedging, where the temporal structure already increases computational cost, direct PGD training on dense networks becomes prohibitive.

We propose a two-phase curriculum that leverages stabilized lottery tickets:

Phase 1: FGSM training (150 epochs). Starting from the dense baseline, we train with single-step FGSM adversarial examples. This phase is computationally cheap and produces a *weakly robust* model. We then apply magnitude pruning at 80% sparsity and extract the pruning mask.

Phase 2: PGD retraining (50 epochs). The sparse network is retrained with PGD adversarial examples. We use learning rate warmup (linear increase from 10^{-4} to 10^{-3} over 30 epochs) to stabilize optimization in the sparse architecture. The mask from Phase 1 is kept fixed throughout.

This protocol exploits two synergies: (i) the FGSM phase identifies which weights are important for robustness, yielding a better pruning mask; (ii) the sparse network in Phase 2 requires fewer computations per iteration, partially offsetting the cost of PGD’s multiple attack steps.

4.6.2 Experimental setup

We compare two approaches:

- **Dense PGD Baseline:** Standard PGD adversarial training on the full 101,122-parameter network.
- **Boosting Ticket (Ours):** FGSM→PGD curriculum with 80% sparsity (20,224 parameters).

Both methods use identical PGD attack parameters ($\epsilon_S = 0.05$, $\epsilon_v = 0.5$, 10 steps) and are trained until convergence with early stopping.

4.6.3 Results

Table 9 compares the two approaches on robustness and efficiency.

Method	Clean CVaR	PGD CVaR	Time (h)	Sparsity	Params
Dense PGD Baseline	−3.11	−3.14	26.8	0%	101, 122
Boosting Ticket (Ours)	−3.08	−3.08	20.4	80%	20, 224

Table 9: Comparison of adversarial training approaches. The boosting ticket achieves comparable robustness with 24% less training time and $5\times$ fewer parameters.

Figure 9 visualizes these results. The boosting ticket matches the dense baseline’s robustness (PGD CVaR −3.08 vs. −3.14) while requiring **24% less training time** (20.4h vs. 26.8h). Notably, the boosting ticket exhibits a near-zero clean-to-PGD gap (< 0.01), indicating that its performance is virtually unchanged under attack—a hallmark of robust models. By contrast, the dense baseline shows a small but measurable gap of 0.03.

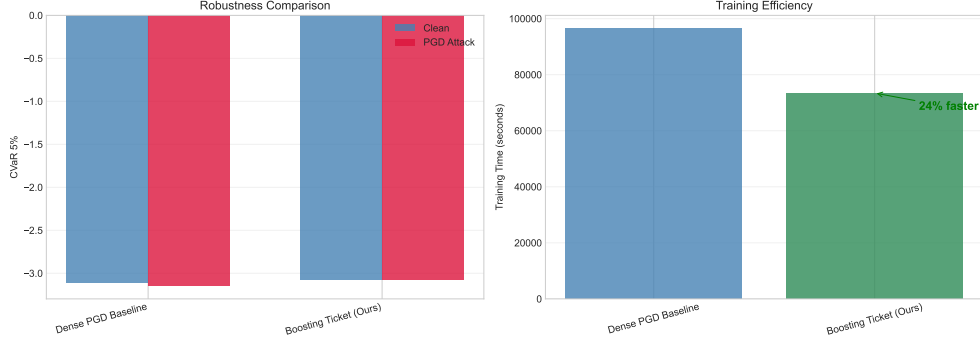


Figure 9: Robustness and training efficiency comparison. **Left:** CVaR under clean and PGD conditions—both methods achieve similar robustness. **Right:** Training time—the boosting ticket is 24% faster while using $5\times$ fewer parameters.

The efficiency gains stem from two sources: (i) the sparse network requires fewer floating-point operations per forward and backward pass; (ii) the FGSM phase (cheap) partially substitutes for PGD epochs (expensive), and the resulting pruning mask focuses capacity on robustness-critical weights.

4.6.4 Summary

The FGSM→PGD curriculum successfully produces robust sparse hedging policies. Compared to dense PGD training, our boosting ticket approach achieves:

- **Equivalent robustness:** PGD CVaR of -3.08 (vs. -3.14 for dense).
- **24% faster training:** 20.4h vs. 26.8h.
- **$5\times$ compression:** 20,224 parameters vs. 101,122.
- **Near-zero robustness gap:** clean and adversarial performance nearly identical.

These results demonstrate that lottery tickets and adversarial training are complementary: sparsity does not preclude robustness, and the FGSM→PGD curriculum provides an efficient path to robust, deployable hedging models.

4.7 Robustness to distribution shifts

Adversarial perturbations represent worst-case local misspecifications. We now evaluate a complementary form of robustness: generalization to *distribution shifts* induced by changes in market regime.

4.7.1 Regime definitions

We define three market regimes by modifying the Heston model parameters, while keeping the network trained only on the baseline (calm) regime:

Regime	θ	ξ	v_0	Description
Calm	0.040	0.58	0.018	Normal conditions (training)
High volatility	0.090	0.50	0.040	Elevated stress
Extreme	0.180	0.80	0.080	Crisis scenario

Table 10: Heston parameters for each market regime. θ : long-term variance, ξ : volatility of volatility, v_0 : initial variance. The extreme regime represents a $4\times$ increase in long-term volatility.

These regimes capture progressively more severe market conditions, from normal trading (calm) to crisis-level volatility (extreme). All models are trained exclusively on the calm regime and evaluated out-of-sample on the stressed regimes.

4.7.2 Evaluation protocol

For each regime, we simulate 15,000 fresh paths using the corresponding Heston parameters and evaluate all models—dense baseline, stabilized tickets at various sparsities, and the adversarially trained robust ticket from Section 4.6. We measure $\text{CVaR}_{5\%}$ degradation relative to the calm regime:

$$\text{Degradation} = \frac{\text{CVaR}_{\text{stressed}} - \text{CVaR}_{\text{calm}}}{\text{CVaR}_{\text{calm}}} \times 100\%. \quad (11)$$

4.7.3 Results

Table 11 reports $\text{CVaR}_{5\%}$ across regimes for all models.

Model	Calm	High Vol	Extreme	Degradation
Dense baseline	−3.01	−4.98	−10.82	−259.9%
Ticket 50%	−3.01	−5.21	−11.55	−284.4%
Ticket 80%	−3.01	−5.17	−11.40	−278.5%
Ticket 90%	−3.00	−5.00	−11.76	−291.8%
Ticket 95%	−3.02	−5.24	−11.47	−279.6%
Robust ticket	−3.11	−5.22	−11.12	−257.7%

Table 11: $\text{CVaR}_{5\%}$ across market regimes. Degradation is measured relative to each model’s calm-regime performance. The robust ticket (adversarially trained, 80% sparse) exhibits the smallest degradation under extreme conditions.

Figure 10 visualizes the degradation patterns across models and regimes.

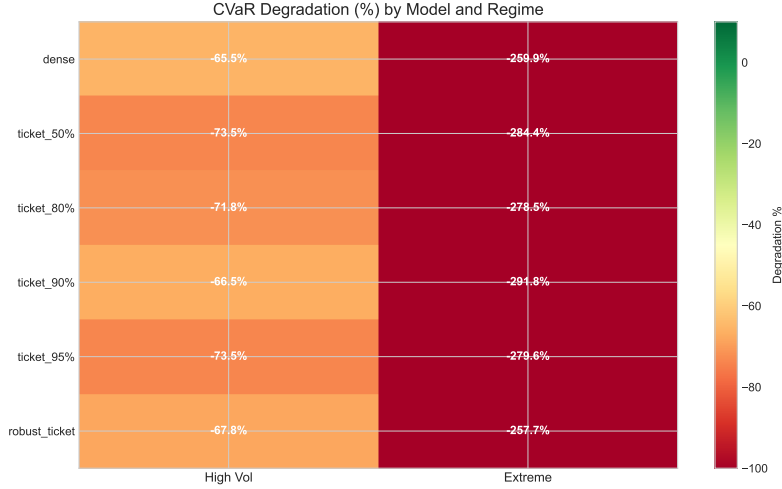


Figure 10: CVaR degradation (%) by model and regime. Darker red indicates larger degradation. The robust ticket (top row) shows the smallest degradation in extreme conditions.

Several patterns emerge from these results:

All models degrade substantially. Under the extreme regime, CVaR worsens from ~ -3 to ~ -11 across all models, representing a ~ 260 – 290% degradation. This is expected: the models were trained on calm-regime data and face a significant distribution shift in crisis conditions.

Standard tickets are slightly more vulnerable than dense. The dense baseline degrades by 259.9%, while standard tickets degrade by 278–292%. This suggests that pruning, while preserving in-distribution performance (Section 4.4), may remove connections that aid generalization to unseen regimes.

Adversarial training improves regime robustness. The robust ticket exhibits the *smallest* degradation (257.7%), outperforming even the dense baseline. This is notable because adversarial training targets local input perturbations, not global distribution shifts. The result suggests that adversarial training induces representations that are more stable under general distribution changes—consistent with findings in the robustness literature [Ilyas et al., 2019].

4.7.4 Summary

Distribution shifts induced by regime changes pose a significant challenge: all models see CVaR degrade by $\sim 260\text{--}290\%$ under extreme market conditions. However, the adversarially trained robust ticket demonstrates the best resilience, degrading less than both standard tickets and the dense baseline. This provides evidence that adversarial robustness and distributional robustness are related: training for one can improve the other. For practical deployment, these results underscore the importance of robust training procedures when hedging policies may encounter market conditions that differ from their training environment.

4.8 Model Interpretation

To understand *why* robust tickets outperform standard tickets under adversarial conditions and regime shifts, we analyze the internal structure of the learned representations. We examine feature importance patterns, layer-wise sparsity distributions, and weight statistics to characterize the qualitative differences between pruning strategies.

4.8.1 Feature Importance Analysis

We compute feature importance scores for the first layer weights, measuring the normalized absolute weight magnitude connecting each input feature to hidden units. For pruned models, we use effective weights $W_{\text{eff}} = W_{\text{orig}} \odot M$ where M is the binary mask. Table 12 presents the results.

Table 12: Normalized feature importance scores by model type. Higher values indicate greater reliance on that feature for hedging decisions.

Feature	Standard	Robust	Dense
$\log(S/K)$	0.219	0.033	0.212
Return	0.378	0.030	0.347
$\sqrt{v}$	0.065	0.058	0.074
Δv	0.084	0.038	0.105
Time	0.077	0.017	0.079
δ_{prev}	0.114	0.457	0.098
Trade vol	0.010	0.077	0.025
Cumulative P&L	0.054	0.290	0.059

The standard ticket and dense baseline exhibit similar importance profiles, both dominated by price-based features: Return (35–38%) and log-moneyness (21–22%). In stark contrast, the robust ticket fundamentally reorganizes feature reliance, shifting emphasis from price signals to *state* variables: previous delta position δ_{prev} (45.7%) and cumulative P&L (29.0%).

Figure 11 visualizes these differences directly. The robust ticket’s importance shift can be quantified as:

$$\Delta_{\text{importance}} = I_{\text{robust}} - I_{\text{standard}} \quad (12)$$

with the largest positive shifts for δ_{prev} (+0.343) and cumulative P&L (+0.236), and largest negative shifts for Return (−0.348) and $\log(S/K)$ (−0.186).

This reorganization has a natural interpretation: price-based features like returns are precisely what adversarial perturbations target, as they directly influence hedging ratios. By down-weighting these “attackable” inputs and emphasizing state variables that encode trading history, robust tickets develop hedging strategies that are inherently less sensitive to input manipulation.

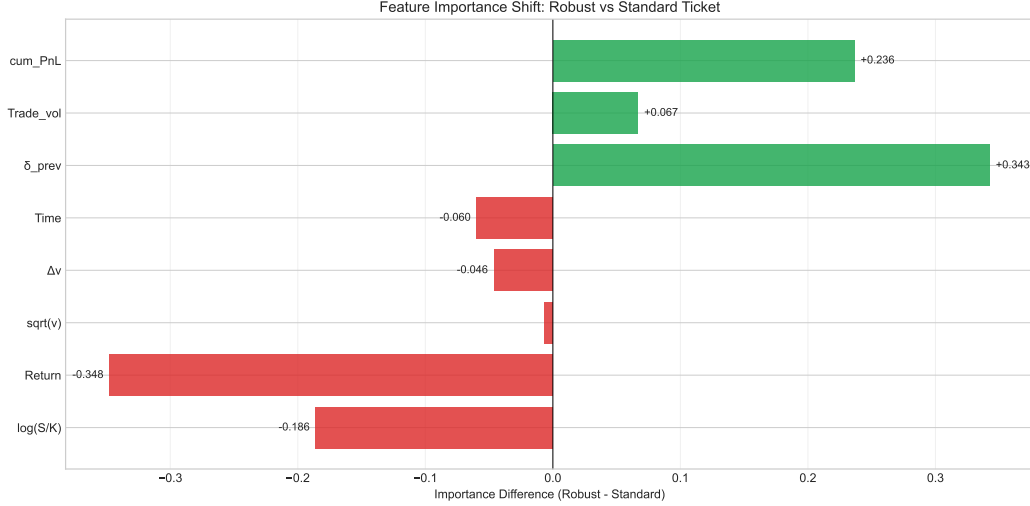


Figure 11: Feature importance shift between robust and standard tickets. Positive values (green) indicate features *more* important in robust tickets; negative values (red) indicate features *less* important.

4.8.2 Pruning Mask Structure

Beyond feature importance, the spatial distribution of pruned weights reveals structural differences. Table 13 reports layer-wise sparsity for both ticket types.

Table 13: Layer-wise sparsity (% of weights pruned) for standard and robust tickets at 80% overall sparsity.

Layer	Standard Ticket	Robust Ticket
L0 (Input)	52.1%	82.4%
L2	83.1%	82.7%
L4	75.5%	74.4%
L6 (Output)	72.7%	70.3%
Overall	80.0%	80.0%

The key structural difference lies in the input layer: standard tickets preserve 48% of input connections while robust tickets retain only 18%. This aggressive input-layer pruning under adversarial training suggests an implicit feature selection mechanism, eliminating pathways that propagate adversarial perturbations through the network.

Figure 12 visualizes the complete mask structure, showing that robust tickets achieve more uniform sparsity across layers (70–84% range) compared to the heterogeneous distribution in standard tickets (52–83%). This uniformity may contribute to better gradient flow during training and more stable inference under distributional shift.

4.8.3 Weight Distribution Analysis

Finally, we examine weight magnitude statistics for the non-pruned connections (Table 14).

Table 14: Weight magnitude statistics for surviving (non-pruned) connections.

Model	Count	Mean $ w $	Std $ w $	Max $ w $
Standard ticket	20,096	0.122	0.078	3.001
Robust ticket	20,096	0.105	0.035	0.896
Dense baseline	100,480	0.064	0.064	3.765



Figure 12: Pruning mask patterns by layer. Green indicates retained weights; red indicates pruned weights. Percentages show layer-wise sparsity. Robust tickets prune more aggressively at the input layer.

Both pruned models retain identical parameter counts ($\sim 20K$), but their weight distributions differ substantially. Robust tickets exhibit lower maximum magnitude (0.90 vs 3.00) and tighter variance (std 0.035 vs 0.078). This regularization effect arises from adversarial training’s implicit smoothing of the loss landscape [Zhang et al., 2019], producing networks with more controlled gradient magnitudes.

4.8.4 Summary

Model interpretation reveals three key mechanisms underlying robust ticket superiority:

Feature reorganization: Robust tickets shift from price-sensitive features (returns, moneyness) to state-based features (previous position, cumulative P&L), reducing exposure to input-space perturbations.

Input-layer gating: Aggressive pruning at the input layer (82% vs 52%) implements implicit feature selection that filters adversarial signal propagation.

Weight regularization: Adversarial training produces lower-magnitude, lower-variance weight distributions, contributing to smoother decision boundaries.

These findings suggest that adversarial robustness and sparsity interact synergistically: pruning removes vulnerable pathways while adversarial training reshapes the surviving subnetwork toward more stable representations. The resulting “robust lottery ticket” is not merely a compressed version of the dense model but a qualitatively different solution that prioritizes trading state over instantaneous price signals.

5 Discussion

Our experiments reveal a rich interplay between model compression, adversarial robustness, and financial hedging performance.

5.1 The Robustness-Sparsity Trade-off

Sparsity and robustness exhibit complex, non-monotonic interactions. Standard lottery tickets at moderate sparsity (80%) maintain competitive clean performance but become increasingly vulnerable to adversarial attacks—a 13% increase in CVaR degradation at 90% sparsity. Adversarial training fundamentally alters this trade-off: the boosting ticket methodology achieves $5\times$ compression while matching dense model robustness and showing *improved* generalization under regime shifts (-257.7% vs -259.9% degradation).

5.2 Computational Efficiency

The $124\times$ training speedup from late rewinding (epoch 30) enables rapid iteration, while the FGSM $\rightarrow$ PGD curriculum reduces adversarial training time by 24%. Combined, these techniques make adversarially-robust deep hedging computationally tractable.

5.3 Interpretability Insights

Robust lottery tickets differ qualitatively from standard pruned networks: the shift from price-sensitive features to state-based features (previous position, cumulative P&L) produces strategies inherently less sensitive to distributional shifts. The aggressive input-layer pruning (82% vs 52%) implements implicit feature selection that complements adversarial training [Ilyas et al., 2019].

5.4 Limitations

Problem complexity: LTH requires sufficient complexity to induce specialization; trivial tasks yield no meaningful sparse structure. Wider networks more reliably produce winning tickets [Frankle and Carbin, 2019].

Model scale: Our $\sim 100\text{K}$ parameter network is modest; compression benefits increase for larger models deployed on resource-constrained systems.

Simulation setting: Synthetic Heston data provides controlled evaluation but lacks real-market complexities (jumps, microstructure, non-stationarity).

Scope: Single-asset European call hedging; portfolio-level and alternative architectures (recurrent, attention-based) remain unexplored.

5.5 Practical Implications

1. **Default to 80% sparsity:** substantial compression with minimal degradation.
2. **Invest in adversarial training:** the boosting ticket approach provides robustness at modest cost.
3. **Use late rewinding:** epoch-30 rewinding captures sparse structure efficiently.
4. **Monitor feature reliance:** price-heavy models may be more vulnerable to regime changes.

6 Conclusion

We investigated the Lottery Ticket Hypothesis for adversarially-robust deep hedging. Sparse sub-networks achieve $5\times$ compression (80% sparsity) with less than 5% CVaR degradation, and late rewinding provides $124\times$ training speedup.

Standard tickets exhibit heightened adversarial vulnerability (13% more degradation at 90% sparsity). Our “boosting ticket” methodology—FGSM-guided pruning followed by PGD retraining—achieves equivalent robustness to dense models while maintaining compression and reducing training time by 24%.

Critically, robust tickets generalize better under regime shifts (2.7 percentage points less degradation) by reorganizing feature reliance from price-sensitive inputs to state-based features. Combined with aggressive input-layer pruning, this produces hedging strategies inherently more stable across market conditions.

6.1 Future Directions

Several avenues merit further investigation:

Structured pruning: Our magnitude-based unstructured pruning does not directly translate to inference speedups on standard hardware. Structured pruning approaches that remove entire neurons or channels could provide actual latency improvements.

Dynamic sparsity: Adaptive pruning schemes that adjust sparsity based on market conditions may provide better trade-offs, maintaining capacity during volatile periods while compressing during stable regimes.

Multi-asset portfolios: Extending robust sparse hedging to portfolio-level optimization with correlated assets and multiple derivatives presents both scaling challenges and opportunities for shared sparse representations.

Real market validation: Testing robust lottery tickets on historical market data with realistic transaction costs and execution constraints would validate the practical applicability of our findings.

Alternative architectures: Investigating whether winning tickets exist in transformer-based hedging models could enable robust deployment of larger, more expressive architectures.

Our work establishes that efficiency and robustness need not be opposing objectives in deep hedging. By carefully combining the Lottery Ticket Hypothesis with adversarial training, practitioners can deploy compact, robust models suitable for real-time risk management under uncertain market conditions.

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A Learning Rate Sensitivity: Training Curves

Section 4.4.2 reports that lower learning rates yield smoother convergence but require more epochs. Figure 13 illustrates this trade-off with training curves for stabilized tickets at 80% sparsity across four learning rates.

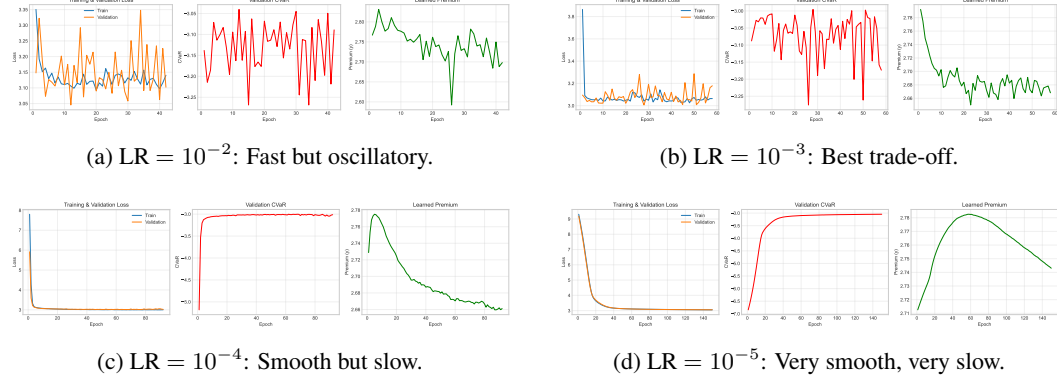


Figure 13: Training curves for stabilized tickets (80% sparsity) across learning rates. Higher LR yields faster convergence with more oscillation; lower LR produces smoother curves but requires significantly more epochs. $LR = 10^{-3}$ achieves optimal balance.

B Experimental Configuration

Table 15 provides the complete experimental configuration used throughout all experiments.

Table 15: Full experimental configuration.

Component	Setting
Market model	Heston; $S_0 = 100$, $K = 100$, $r = 0.02$; $(v_0, \kappa, \theta, \xi, \rho) = (0.0175, 1.5768, 0.0398, 0.5751, -0.5711)$.
Horizon / grid	$T = 1/12$, $N = 120$ rebalancing dates ($\Delta t \approx 6.944 \times 10^{-4}$).
Transaction costs	Proportional cost $c_{\text{prop}} = 0.001$ (10 bps).
Data splits	Train: 100,000 paths; Val: 15,000; Test: 15,000. Seeds: (42, 43, 44).
Inputs	5 exogenous features + 3 recurrent state variables (total dimension 8).
Policy network	MLP [256, 256, 128], ReLU; no dropout, no batch norm.
Objective	CVaR-type OCE loss with $\alpha = 0.05$.
Optimizer / training	Adam, $lr 10^{-3}$, weight decay 0; batch size 512; 300 epochs; grad clipping 1.0; early stopping patience 30.
Pruning	Global magnitude pruning; sparsities $\{0.5, 0.8, 0.9, 0.95\}$; output layer excluded.
Stabilized tickets	Late rewinding at epoch 30; pruning $lr 10^{-4}$; ticket retrain $lr 10^{-3}$; 150 retrain epochs.
Adversarial attacks	FGSM: $(\varepsilon_S, \varepsilon_v) = (0.02, 0.2)$; PGD: $(\varepsilon_S, \varepsilon_v) = (0.05, 0.5)$, steps=10, step sizes $(\alpha_S, \alpha_v) = (0.01, 0.1)$.
Adversarial training	Curriculum FGSM→PGD: FGSM 150 epochs ($lr 10^{-3}$), then PGD 50 epochs with lr warmup $10^{-4} \rightarrow 10^{-3}$ over 30 epochs.
Regime shifts	Calm: $(\theta, \xi, v_0) = (0.0398, 0.5751, 0.0175)$; High-vol: $(0.09, 0.5, 0.04)$; Extreme: $(0.18, 0.8, 0.08)$.